

Introduction to Social Data Analytics: Week 4 Part 2

Achievement Gaps

- ▶ The motivation for this paper is that many students in India are not at their grade level in terms of understanding
- ▶ If students are not at their grade level, they may not actually learn much from attending school
- ▶ In this class we will document that (1) on average students are performing well below grade level and (2) a year in school is not associated with an additional year of understanding

Variables

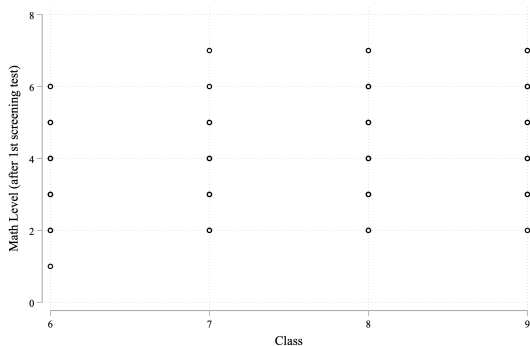
- ▶ The variables we will be exploring are in the dataset `mindspark_levels.dta`
 - ▶ `class` – the grade the student is enrolled in
 - ▶ `mathelevel` – the assessed level of the student's understanding
- ▶ To begin, let's provide a scatter plot of assessed math level on actual grade
- ▶ First step: load the data

```
cd "~/Dropbox/Teaching/EP5/online/04_week/data"  
use mindspark_levels.dta
```

Basic Scatterplot

- The most basic scatterplot we could create from this data is displayed below

```
/* set scheme of graph */  
set scheme plotplainblind  
/* construct figure */  
twoway scatter mathlevel class
```



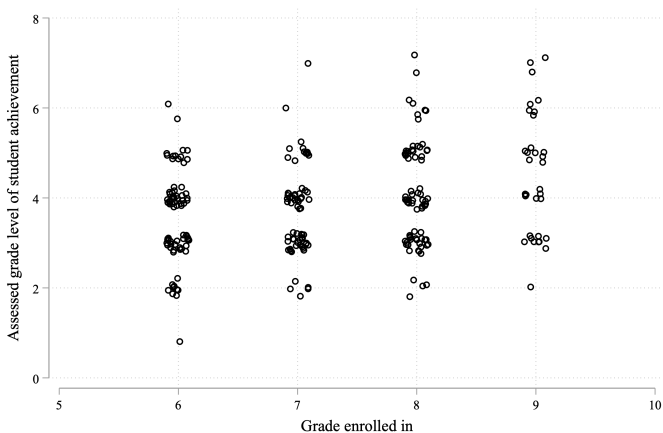
- ▶ That graph didn't seem particularly informative. . .
- ▶ The reason is because so many points are overlapping
- ▶ There are likely many individuals represented by each dot, but from this graph it's not possible to tell how many
 - ▶ For example, we know there are individuals in Grade 6 that are being evaluated as having grade 1-6 level in math, but we don't know how many are in each category
- ▶ One way we can improve the graph is to use the `jitter()` option

The `jitter()` option

- ▶ Scatter plots give us a transparent way to view our data
- ▶ But when variables are categorical (like `grade` and `mathlevel` in our example), they can be less informative because so many observations overlap
- ▶ `jitter()` adds a small amount of noise (small perturbations) to the data so that the points no longer overlap on the graph
 - ▶ In parentheses, you place a number that determines how much noise is added. (5) is a common choice. A value of (100) would make it so that the graph is now completely dominated by noise

Improved Scatterplot with Categorical Data

```
twoway scatter mathlevel class, jitter(5) ///  
  xlabel(5(1)10) ///  
  xtitle("Grade enrolled in") ///  
  ytitle("Assessed grade level of student achievement")
```



Takeaway so Far

- ▶ As you can see in the figure, very few students are performing at their grade level, with none performing above grade level.
- ▶ Large clusters at grades several years below grade level
 - ▶ For example, for students in grade 6, the largest clusters are at 4th grade math level and 3rd grade math level
- ▶ This gives us a sense of the scope of the issue. Students understanding is well below their grade level

Learning

- ▶ Next, let's try to get a sense of how learning is evolving over time
- ▶ Two potential “stories”
 - ▶ Students are below grade level, but each year they are still learning (i.e. increasing grade by 1 is associated with a 1-year increase in assessed math level)
 - ▶ Students performing below grade level makes learning difficult (i.e. increasing grade by 1 is associated with less than a 1-year increase in assessed math level)
- ▶ We can study this question in a regression framework

Regression

- ▶ We are interested in the relationship between assessed math level and grade

$$\text{Math Level}_i = \alpha + \beta \cdot \text{Grade}_i + \varepsilon_i$$

- ▶ β tells us how much we expect assessed math level to increase for a 1-unit increase in Grade
- ▶ If this is much lower than 1, then students will fall farther and farther behind each year

Interpreting the Output

- ▶ $\hat{\beta} = 0.293$
- ▶ Interpretation: A 1-unit increase in grade is associated with only a 0.293 year increased in assessed math level
- ▶ In other words, students are learning over time, but at a much lower rate than is envisioned by the curriculum
 - ▶ Ideally, 1 year of school should be associated with an additional year of assessed level

Need for Intervention

- ▶ We have documented a few things in our data
 - ▶ 1. Students (on average) are assessed to be performing below grade level
 - ▶ 2. There is large variation in assessed grade level within a given grade
 - ▶ 3. Learning over time is slower than intended
- ▶ Maybe an intervention that “Teaches at the Right Level” can help alleviate some of these issues

Life Expectancy and Annual Income

- ▶ In a previous example, we explored the association between annual income and life expectancy through a linear regression model

$$\text{Life Expectancy}_i = \alpha + \beta \text{Annual Income}_i + \varepsilon_i$$

- ▶ In this example, both Y (Life Expectancy) and X (Annual Income) are continuous variables
 - ▶ Continuous means they can take on any value in a given range
- ▶ The interpretation of β is the expected increase in Life Expectancy for a 1-unit change in annual income
- ▶ Let's talk about how to interpret models in which either Y or X (or both) are binary indicator variables

Binary Outcome, Continuous Explanatory Variable

- ▶ Let's imagine we are interested in whether temperature impacts voting
- ▶ We have an individual-level dataset that indicates whether an individual voted ($\text{Vote}_i = 1$ if voted and zero otherwise), as well as the temperature in Fahrenheit in the location they voted
- ▶ We propose modeling this via a linear regression framework

$$\text{Vote}_i = \alpha + \beta \cdot \text{Temperature}_i + \varepsilon_i$$

Estimates

- ▶ Imagine we find $\alpha = 0.1$ and $\beta = 0.01$. How do we interpret these numbers?
- ▶ Before we said the interpretation of the slope coefficient is “the expected increase in Y for a 1-unit change in X ”

Estimates

- ▶ Imagine we find $\alpha = 0.1$ and $\beta = 0.01$. How do we interpret these numbers?
- ▶ Before we said the interpretation of the slope coefficient is “the expected increase in Y for a 1-unit change in X ”
- ▶ We expect our variable *Vote* to increase by 0.01 if Temperature increases by a degree
- ▶ A 0.01 increase in a binary indicator implies a 1 percentage point increase in the probability the value of the variable is equal to 1
 - ▶ In this case, we expect voting to increase by 1 percentage point for every degree increase in the temperature

Interpretation Through Predicted Values

- ▶ The easiest way to think about this may be through forming predicted values
- ▶ What is the predicted value of *Vote* if the temperature is 50 degrees

$$\text{Predicted Vote} = 0.1 + 0.01 * 50 = 0.6$$

Interpretation Through Predicted Values

- ▶ The easiest way to think about this may be through forming predicted values
- ▶ What is the predicted value of *Vote* if the temperature is 50 degrees

$$\text{Predicted Vote} = 0.1 + 0.01 * 50 = 0.6$$

- ▶ What is the predicted value of *Vote* if the temperature is 51 degrees

$$\text{Predicted Vote} = 0.1 + 0.01 * 51 = 0.61$$

- ▶ So increasing the temperature by 1 degree is expected to increase the probability of voting by 1 percentage point

Continuous Outcome, Binary Explanatory Variable

- ▶ Now we are interested in the effect of a Universal Basic Income (UBI) payment on total earnings of individuals
 - ▶ There are now a few experiments on UBI that are either complete or in progress
- ▶ In an experimental setup, we could model earnings as a function of whether you are given access to the UBI payment (i.e. UBI is binary)

$$\text{Total Earnings}_i = \alpha + \beta \cdot \text{UBI} + \varepsilon_i$$

Interpretation through Predicted Values

- ▶ What is the predicted earnings for someone who does not receive UBI

$$\text{Predicted Earnings} = \alpha + \beta \cdot 0 = \alpha$$

Interpretation through Predicted Values

- ▶ What is the predicted earnings for someone who does not receive UBI

$$\text{Predicted Earnings} = \alpha + \beta \cdot 0 = \alpha$$

- ▶ What is the predicted earnings for someone who does receive UBI

$$\text{Predicted Earnings} = \alpha + \beta \cdot 1 = \alpha + \beta$$

- ▶ So β is the difference in earnings for participants that receive UBI, relative to those that do not

Binary Outcome Binary Explanatory Variable

- ▶ Let's change the prior example slightly. Instead of studying the impact of UBI on earnings, let's study the impact on employment

$$\text{Employment}_i = \alpha + \beta \cdot \text{UBI} + \varepsilon_i$$

- ▶ Now both Y and X are binary

Interpretation through Predicted Values

- ▶ What is the predicted value of employment for someone who does not receive UBI

$$\text{Predicted Employment} = \alpha + \beta \cdot 0 = \alpha$$

- ▶ In other words, of those who do not receive UBI, α percent are employed
- ▶ What is the predicted earnings for someone who does receive UBI

$$\text{Predicted Employment} = \alpha + \beta \cdot 1 = \alpha + \beta$$

- ▶ In other words, of those who do not receive UBI, $\alpha + \beta$ percent are employed
- ▶ β is the percentage point difference in employment rates between those who receive UBI vs. those that do not

The Impact of Treatment on Test Scores

- ▶ We want to understand the impact of treatment (being offered free Mindspark) impacts test scores
- ▶ We will estimate this impact in a regression framework

$$\text{Endline Test Score}_i = \alpha + \beta \cdot \text{Treat}_i + \varepsilon_i$$

- ▶ Endline Test Score_{*i*} ranges from 0 to 1 and represents the percent of questions that are correct
- ▶ Treat_{*i*} is a binary indicator that is 1 if the individual is in the treatment group and zero otherwise

Does Treatment Impact Endline Math Scores

```
cd "/Users/davidarnold/Dropbox/Teaching/EP5/online/04_week/data"  
use mindspark_data.dta, clear
```

```
/Users/davidarnold/Dropbox/Teaching/EP5/online/04_week/data
```

```
reg per_math2 treat
```

Source		SS		df		MS		Number of obs	=	539
-----+-----								F(1, 537)	=	28.72
Model		.813404311		1		.813404311		Prob > F	=	0.0000
Residual		15.20668		537		.02831784		R-squared	=	0.0508
-----+-----								Adj R-squared	=	0.0490
Total		16.0200843		538		.029777108		Root MSE	=	.16828

per_math2		Coefficient		Std. err.		t		P> t		[95% conf. interval]
-----+-----										
treat		.0777008		.0144978		5.36		0.000		.0492214 .1061802
_cons		.4647551		.0101847		45.63		0.000		.4447484 .4847619

Interpretation

- ▶ We found that the slope coefficient $\hat{\beta} = 0.078$
- ▶ Our general formula for interpreting slope coefficients: a 1-unit change in `treat` is expected to increase `per_math2` by 0.078
- ▶ In plain (interpretable) English: Treated students that were offered free Mindspark tuition saw a 7.8 percentage point increase in math scores relative to control students
 - ▶ Concept check: What is the average score for the control group at endline?

Does Treatment Impact Endline Hindi Scores

```
reg per_hindi2 treat
```

Source	SS	df	MS	Number of obs	=	539
-----+-----				F(1, 537)	=	16.12
Model	.572530953	1	.572530953	Prob > F	=	0.0001
Residual	19.076815	537	.035524795	R-squared	=	0.0291
-----+-----				Adj R-squared	=	0.0273
Total	19.6493459	538	.036522948	Root MSE	=	.18848

per_hindi2	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
treat	.0651886	.0162382	4.01	0.000	.0332904	.0970868
_cons	.5200244	.0114073	45.59	0.000	.4976159	.5424329

Interpretation

- ▶ We found that the slope coefficient $\hat{\beta} = 0.065$
- ▶ Our general formula for interpreting slope coefficients: a 1-unit change in `treat` is expected to increase `per_hindi2` by 0.065
- ▶ In plain (interpretable) English: Treated students that were offered free Mindspark tuition saw a 6.5 percentage point increase in math scores relative to control students

Conditional Regressions

- ▶ So far we have estimated the average effect across all students
- ▶ Often times it is informative to consider **heterogeneity** in treatment effects (i.e. how treatment effects vary across groups)
- ▶ One way to test for heterogeneity in treatment effects is to estimate a series of conditional regressions
- ▶ We will estimate our primary regression but separated by
 - ▶ Sex
 - ▶ See textbook for examples that split by socioeconomic status and baseline learning

Conditional Regressions

- ▶ The way to estimate a conditional regression is to type

```
reg Yvar Xvar if (logical statement)
```

- ▶ We will replace (logical statement) with something that specifies a given subset of the data
 - ▶ For example, if we would like to restrict to females, we would type

```
if st_female1==1
```

```
/* regression of endline test scores on treatment status for females */
reg per_math2 treat if st_female1==1
```

Source		SS	df	MS	Number of obs	=	414
-----+-----					F(1, 412)	=	17.78
Model		.518287614	1	.518287614	Prob > F	=	0.0000
Residual		12.0091958	412	.029148533	R-squared	=	0.0414
-----+-----					Adj R-squared	=	0.0390
Total		12.5274834	413	.03033289	Root MSE	=	.17073

-----+-----						
per_math2		Coefficient	Std. err.	t	P> t	[95% conf. interval]
-----+-----						
treat		.0707653	.016782	4.22	0.000	.0377763 .1037543
_cons		.4636151	.011838	39.16	0.000	.4403447 .4868854
-----+-----						

```
/* regression of endline test scores on treatment status for males */
reg per_math2 treat if st_female1==0
```

Source		SS	df	MS	Number of obs	=	125
-----+-----					F(1, 123)	=	12.67
Model		.323197879	1	.323197879	Prob > F	=	0.0005
Residual		3.13679016	123	.025502359	R-squared	=	0.0934
-----+-----					Adj R-squared	=	0.0860
Total		3.45998804	124	.027903129	Root MSE	=	.15969

-----+-----						
per_math2		Coefficient	Std. err.	t	P> t	[95% conf. interval]
-----+-----						
treat		.1017787	.0285899	3.56	0.001	.0451867 .1583707
_cons		.4684034	.0198077	23.65	0.000	.4291953 .5076114
-----+-----						

Interpretation

- ▶ $\hat{\beta}^{female} = 0.071$
 - ▶ Treatment increases percent score correct by 7.1 percentage points for females
- ▶ $\hat{\beta}^{male} = 0.102$
 - ▶ Treatment increases percent score correct by 10.2 percentage points for males
- ▶ Treatment effect larger for males, but large improvements in performance for both female and male students

Conclusion

- ▶ Using technology to “Teach at the Right Level” seems like a promising intervention in places in which there is tremendous heterogeneity in baselines levels of understanding
- ▶ There is potential for even better educational design!
 - ▶ Maybe the best intervention would blend Mindspark with standard teaching
 - ▶ Giving teachers more information about the level of their students could aid instruction in the classroom
 - ▶ This automated aspect of the software could free up teacher time for other activities that also improve student performance